

Homework (Carolina Reaper)

- Q1.** A music track has a rhythm pattern of 3 : 4 between the guitar and drums. If the guitar plays 12 notes, how many notes will the drums play?
(A) 12
(B) 16
(C) 20
(D) 24
(E) 28
- Q2.** An artist mixes paints in a ratio of 2 : 5 for a specific shade. If she uses 10 mL of the first paint, how much of the second paint should she use?
(A) 20 mL
(B) 25 mL
(C) 30 mL
(D) 35 mL
(E) 40 mL
- Q3.** A sound wave has a frequency ratio of 1 : 2 between two notes. If the first note has a frequency of 20 Hz, what is the frequency of the second note?
(A) 20 Hz
(B) 40 Hz
(C) 60 Hz
(D) 80 Hz
(E) 100 Hz
- Q4.** A canvas has a width to length ratio of 3 : 5. If the width is 9 inches, what is the length of the canvas?
(A) 12 inches
(B) 15 inches
(C) 18 inches
(D) 21 inches
(E) 24 inches
- Q5.** A musician plays a rhythm with a 2 : 3 ratio between quarter notes and eighth notes. If she plays 8 quarter notes, how many eighth notes will she play?
(A) 8
(B) 12
(C) 16
(D) 20
(E) 24
- Q6.** An art piece has a color ratio of 4 : 7 between two shades. If 16 parts of the first shade are used, how many parts of the second shade should be used?
(A) 20
(B) 24
(C) 28
(D) 32
(E) 36
- Q7.** A sound system has a volume ratio of 2 : 3 between two speakers. If the first speaker is set to 6, what should the second speaker be set to?
(A) 8
(B) 9
(C) 10
(D) 12
(E) 15
- Q8.** A painter uses a 3 : 2 ratio of paint to thinner. If she uses 6 mL of paint, how much thinner should she use?
(A) 2 mL
(B) 3 mL
(C) 4 mL
(D) 6 mL
(E) 8 mL

Open Ended Questions (FRQ)

- Q9.** A musician is creating a song with a ratio of 5 : 6 between the duration of the intro and the outro. If the intro lasts 15 seconds, how many seconds will the outro last? Show your work and explain your reasoning.
- Q10.** An artist is designing a mural with a ratio of 4 : 9 between the width and length. If the width is 12 feet, what will be the length of the mural? Use cross multiplication to solve the proportion and explain your answer.

Chef's Tips

- Check for unit conversions
- Look for correct proportion setup
- Verify student explanations